

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	07 July 2026
Team ID	
Project Name	Visualization tool for electric vehicle charge and range analysis
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Import	Import EV dataset (Excel/CSV)Validate dataset format
FR-2	Data Processing	Clean missing valuesPrepare data for visualization
FR-3	Dashboard Visualization	Display interactive charts and graphsShow KPIs such as charging time, battery capacity, and driving range
FR-4	Data Filtering	Filter data by vehicle model, manufacturer, year, battery capacity, etc
FR-5	Data Analysis	Compare EV charging time, battery capacity, and driving rangeIdentify trends and patterns
FR-6	Report & Insights	Generate visual insights from the dashboardEnable users to interpret EV performance effectively

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Dashboard should be simple, interactive and easy to use.
NFR-2	Security	Only authorized users can modify the dataset.
NFR-3	Reliability	Dashboard should display accurate and consistence information
NFR-4	Performance	Dashboard should load charts and filters quickly
NFR-5	Availability	Dashboard should be accessible whenever required
NFR-6	Scalability	Dashboard should support larger EV datasets in the future